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Sensing the World Reflection 2:

The last two weeks have been enlightening for me as I knew extraordinarily little about computer and electrical engineering, which is what we focused on. In the first class of this part of the course, we received an Arduino Nano ESP32 which we connected to our computers and then created simple programs in python to run. This included outputting our name or the word hello. Along with this we learned basic python knowledge such as what is a comment, function, string, and variable, as well as essential functions that we would be utilizing. In the next class we worked on creating circuits. We received a solderless breadboard, wires, resistors, a potentiometer, and LEDs. We then created simple circuits using these. We got the power by connecting to the power supply which was DC and used a voltage of 5 V. Then we experimented with a switch to turn our LED on and off. Then we used our materials provided plus the Nano ESP32 to reimagine the security system example provided in an early class. We connected the Arduino to the breadboard and then connected resistors and wires to different pins of the Arduino. Finally, we used ChatGPT to create the code which the system would run off and after copying that into the program we were able to make the system function how we intended. In the class after that we continued utilizing much of the same tools. We started with something new, a servo motor which we needed to use to get an arm attached to rotate 180 degrees when we pressed a switch. This involved connecting to the Arduino and using ChatGPT to create a program for us that would rotate the arm when we pressed the switch. We needed to make sure that the right wires connected to the right pins so that everything would work as intended. Then we created a set-up where we had a moisture sensor connected to the breadboard and we used the DC supply to see how the voltage changed depending on the moisture level. We did this manually first and then again, but we used ChatGPT to create code which would take readings of the voltage at different moisture levels automatically. Finally, we then performed data analysis on both the manual and automatic data to see any differences.

I learned skills in both electrical and computer engineering during this section of the course. I do not have much if any prior knowledge with these two fields, so the first few classes were confusing, even stuff we did the last day did not click right away. One of the important things was how to connect the wires so that everything worked properly. Originally, we could not get the pot and LED to work because we had misplaced one of the wires connected to the pot, accidentally putting it in the first slot instead of the second slot. Once we fixed this the pot worked as intended. Something similar happened with the Servo Motor where we could not figure out why our program was not working. It turns out

we just had not grounded the circuit properly so after connecting it to the ground it functioned as intended with no hiccups. I also learned about how useful AI tools are in helping to write code. Honestly if we could not use them, I would not have known how to do anything but with the help of these tools we could create code that ran the program exactly how we wanted it to. The major learning curve here was the prompt that we gave AI for the sensor program. When we told the AI to create a program that would automatically read the voltage output, we forgot to put in how many times we wanted the program to take a reading. So, it kept taking readings even after we had the ten we needed. This made us realize that we had to be more specific about how we wrote our program and once we specified that we only needed ten readings the program began functioning how we wanted it to. I still need to learn about electrical and computer engineering, but this portion of the course helped me to have at least a basic understanding of what the key concepts of these disciplines are.